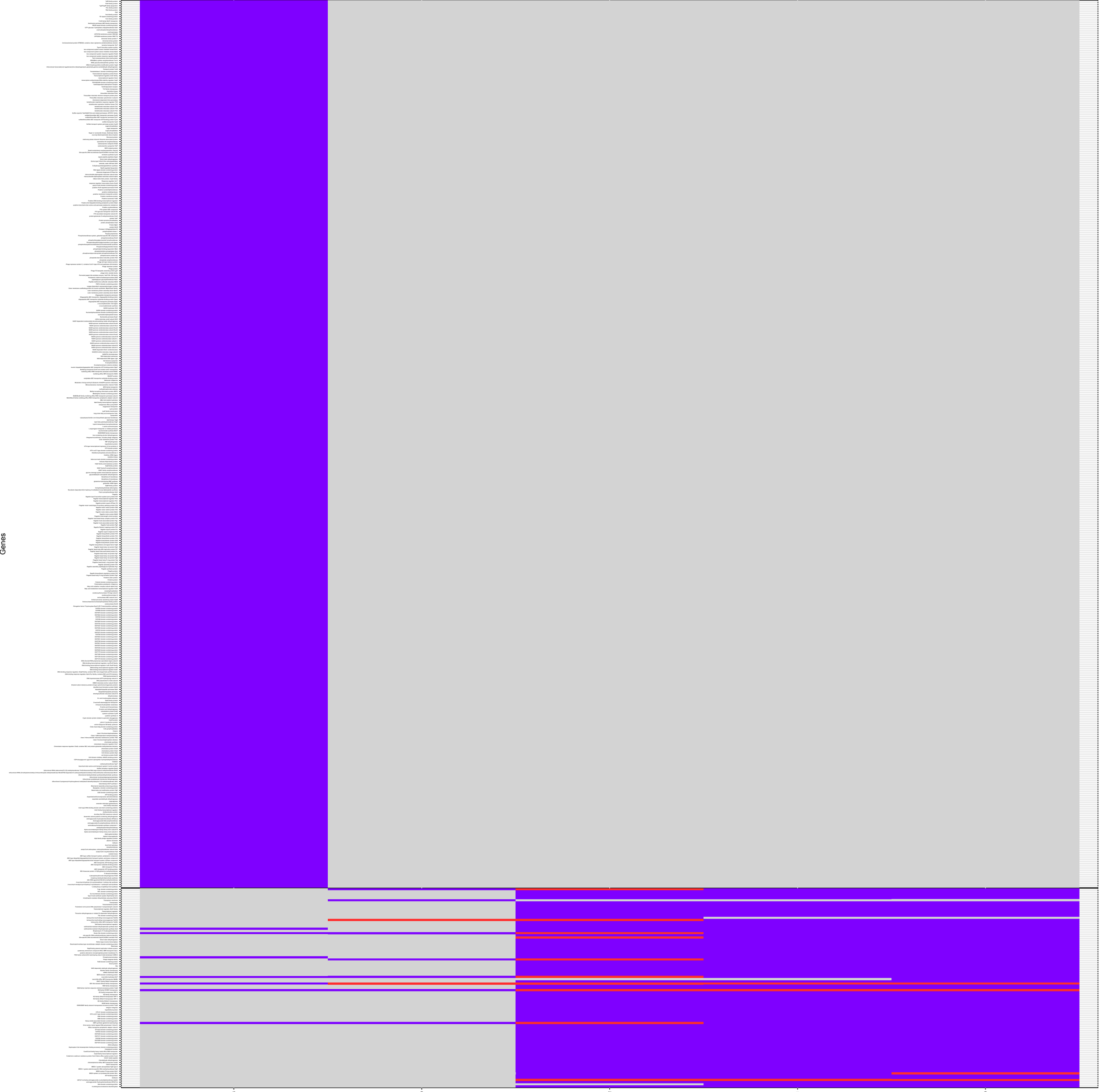
Genes
TpiR family protein
Cupin domain protein
TyrR/TrpR family transcriptio
FNU family protein
FNU family protein
HigA
YecA family protein
YD repeat protein
YD repeat protein
TrmB family, MraW transcriptor
Bacillibactin permease (MFS family) transporte
MOSHE-typical domain-containing protein
LTTR-glycine-N-acetyltransferase fusion protein
uvrA2 phosphotriesterase/methylase
smaC protease
UPF0756 membrane protein
UPF0056 membrane protein P96CY6
Uncharacterized protein STM3048, contains class I glutamine amidotransferase domain
tyrosine transporter TyrP
Type III restriction enzyme, res subunit
two-component system sensor histidine kinase KinC
two-component system sensor histidine kinase BaeS
two-component system response regulator BaeR
two-component system response regulator ArcA
Tat/ twin-arginine translocation pathway small protein
Tat/ twin-arginine translocation pathway small protein
tRNA (adenine(22)-N(1))-methyltransferase TrmK
rRNA 2-O-ribosylguanosine modification protein RsgG
trifunctional transcriptional regulator/proline dehydrogenase/L-glutamate gamma-semialdehyde dehydrogenase
Transport protein TolB
Translation-associated GTPase
Transcriptional regulatory protein AtoA
Transcriptional regulator, AraC family
Transcriptional regulator, AraC family
transcription antitermination/RNA-binding regulator CspD
TPR-MEJ036 domain-containing protein
TonB-dependent receptor, plug
TonB-dependent receptor
TrmJ family transcriptase
Ribosomal protein L7Ae
Thiosulfate reductase electron transport protein phsA
Thiosulfate reductase, subunit PhsB
Homeobox-dependent kinA pseudokinase
tetrathionate response regulator TorR
tetrathionate reductase subunit TorA
tetrathionate reductase subunit TorC
tetrathionate reductase subunit TorD
tetrathionate reductase subunit TorA
SurfAs exporter TauE/SafE/YfcA and related permeases, UPF0721 family
sulfate/thiosulfate ABC transporter permease CysT
sulfate/thiosulfate ABC transporter permease CysT
sulfate/thiosulfate ABC transporter ATP-binding protein CysA
sulfate/thiosulfate ABC transporter ATP-binding protein CysA
SurfAs transport system permease protein CysW
sugar-phosphatase
sugar-phosphatase
sugar-phosphatase
Sugar or nucleoside kinase, ribokinase family
transcript attenuation/antitermination protein NusG
Ribosomal protein
stationary-phase-induced ribosome-associated protein
sulfoquinovosidase sulfoquinase
sulfoquinovosidase sulfoquinase SmaB
multicomponent sulfoquinase SmaC
Small-conductance mechanosensitive channel
Site-specific DNA-recombinase SpoIVCA/SKI family
virulence synthase CysA
virgB proteins (pertactin) CysA
Serine-type D-Ala-D-Ala carboxypeptidase
ketamine, water reformase SurE
S-adenosylmethionine synthase MetK
RpoD-regulated lipoprotein
RNA ligase domain-containing protein
ribosomal protein S4 delta pseudouridylase
ribonucleotide diphosphate reductase subunit beta
ribonucleotide diphosphate reductase subunit alpha
ribose-methyl transferase RsmJ
Response regulator GcrY
response regulator transcription factor RcaB
response regulator transcription factor RcaB
putative PrsW-regulated peptidase PrsW
Putative nucleotidyltransferase
putative membrane transporter
putative membrane transporter protein
Putative membrane protein
Protein RNA-binding regulator RsmA
Putative DNA-binding transcriptional regulator
Putative ELD-dependent binding periplasmic protein RbpA
putative branched-chain amino acid permease (isoleucine-selective)
Putative acyltransferase
PTS glucose transporter subunit IIA
PTS glucose transporter subunit IIA
PTS mannitol transporter subunit IIC
protein-glutamate O-methyltransferase CheR
protein YqiF
Protein tyrosine phosphatase
protein phosphatase YfkJ
Protein RgdI
protein BarB
Promoter 6-like protein
phosphorylase kinase 1
Polyrib polymerase
Phosphotransferase system, galactitol-specific IIB component
phosphotransferase RcsD
phosphorribosylglycinamide formyltransferase
phosphoribosylformylglycinamidine synthase
phosphoribosylaminoimidazolesuccinocarboxamide synthase
Phosphoseryltyrosylphosphotransferase
phosphorylcholine binding lipoprotein MlaA
phosphorylcholine phosphatase EutA
phosphotransferase system, mannitol-specific IIA component
phosphotransferase protein Ser
phosphatase-like protein (hexulose-6-phosphate synthase)
Phage-related tail protein, putative
Phage tail tape measure protein
Phage repressor protein C, contains CroC1-type HTH and peptidase s24 domains
Phage protein
Phage protein
Phage P5 baseplate assembly protein gp7
Phage protein
Perchlorat pepsin-like amidase enzyme, TspA7/YiiX, CAP family
Periplasmic dipheno/indophenoloxidase ElbA
Peptidyl-tRNA hydrolase, bacterial type PTH1
Peptidyl-methionine sulfoxide reductase MsrB
PAPS 3'-domain-containing protein
oxygen-dependent coproporphyrinogen-III oxidase
Outer membrane scaffolding protein for invasin synthesis, MlaA/YhcN/Pqi4 family
outer membrane protein assembly factor BamC
outer membrane protein assembly factor BamB
Oligopeptide transporter permease
Oligopeptide ABC transporter, periplasmic substrate-binding protein OppA
oligopeptide ABC transporter substrate-binding protein OppA
oligopeptide ABC transporter permease OppC
oligopeptide ABC transporter permease OppC
4-vinylreductase/vinylenase synthase
NUDIX hydrolase YhcG
NUDIX domain-containing protein
Nucleotidyltransferase domain-containing protein
nucleoside diphosphatase kinase
NTP pyrophosphohydrolase MazG
nitrite reductase small subunit NirD
NAD(P)-dependent oxaloacetate decarboxylating malate dehydrogenase
NADH-quinone oxidoreductase subunit NuoN
NADH-quinone oxidoreductase subunit NuoM
NADH-quinone oxidoreductase subunit NuoL
NADH-quinone oxidoreductase subunit NuoK
NADH-quinone oxidoreductase subunit NuoJ
NADH-quinone oxidoreductase subunit NuoI
NADH-quinone oxidoreductase subunit NuoH
NADH-quinone oxidoreductase subunit NuoG
NADH-quinone oxidoreductase subunit NuoF
NADH-quinone oxidoreductase subunit NuoE
NADH-quinone oxidoreductase subunit NuoC
NADH-quinone oxidoreductase subunit NuoB
NADH-quinone oxidoreductase subunit J
NADH-quinone oxidoreductase subunit CII
NADH-quinone oxidoreductase subunit B
NADH-quinone oxidoreductase subunit A
NADH-dependent flavin oxidoreductase
NAD(P)H-nitrite reductase, large subunit
NAD-dependent epimerase
NAD-dependent epimerase
NAD-dependent DNA ligase LigA
N-acetylmuramoyl-L-alanine amidase
N-acetylmannosamine-6-phosphate 2-epimerase
N-acetylmuramoyl-L-alanine amidase
murein tripeptide/tripeptide ABC transporter periplasmic substrate-binding protein
Multidrug transporter EmrE and related cation transporters
multidrug efflux RND transporter permease subunit MdtC
multidrug efflux RND transporter permease subunit MdtB
multidrug efflux MFS transporter
MnhG/PhaG subunit
molybdate ABC transporter substrate-binding protein
Modulator of drug activity B (Colicin V/YNFP) (cytosine inducible)
Monooxoclavaldine mechlavaminidine cleavel YbdG
Monooxoclavaldine mechlavaminidine cleavel
MFS family transporter
Methyl-accepting chemotaxis protein (MCP)
Metallophos domain-containing protein
MmdA/MtrB family multidrug efflux RND transporter permease subunit
MmdA-MtrA-MtrB family multidrug efflux RND transporter membrane fusion protein
MBL fold metallo-hydrolase
MarR family transcriptional regulator
Major facilitator superfamily MFS_1
magnesium transporter
Lysin protein
LysM domain protein
long-chain-fatty-acid transporter FadL
Lipocalcin
Lipoprotein/cell/cantrile core transglutaminase glycosyltransferase
lipid kinase YegS
lipid A-linked palmitoyltransferase PagP
Lipid A-linked palmitoyltransferase PagP
L-serine ammonia-lyase
L-asparagine transporter or related permease
Iron-containing alcohol dehydrogenase
ISSod3/IS903 family transposase
iron-containing alcohol dehydrogenase
Integrase/recombinase, includes phage integrase
Integrase/recombinase, includes phage integrase
INF dehydrogenase
hypothetical protein
HTH-type transcriptional repressor of iron proteins A
HTH-type transcriptional repressor of iron proteins A
HTH cro/C1-type domain-containing protein
Histidinol-phosphate aminotransferase 2
histidine kinase
histidine kinase
Helix-turn-helix domain-containing protein
Helix-turn-helix domain protein
HAMP family, and transducer protein
GspB family protein
GNAT family acetyltransferase
GNAT family acetyltransferase
glycine cleavage system transcriptional repressor
glycerophosphoryl diester phosphodiesterase
Glutathione S-transferase
Glutathione S-transferase
glutaminase-hydrolyzing GMP synthase
glucose-1-dehydrogenase
GlpM family protein
formyltransferase/hydrolase subunit/class
Ferredoxin-dependent (E)-4-hydroxy-3-methylbut-2-enyl diphosphate synthase
Fimbrial assembly protein (PilN)
Flagellin
Regulator type (K-secretion system gene protein FlgP
Regulator transcriptional regulator FliA1
Regulator transcriptional regulator FliA1
Regulator protein export ATPase FlhA
Flagellar motor switch/secretion protein FliN
Regulator motor switch protein FliM
Regulator motor switch protein FliG
Regulator motor protein MotB
Regulator motor protein MotA
Flagellar hook-length control protein
Regulator hook-associated protein FlgL
Regulator hook-associated protein FlgL
Regulator hook-associated protein FlgK
Regulator hook protein FlgE
Regulator forward-capping protein FliD
Regulator export protein FliJ
Regulator export chaperone FliS
Regulator export chaperone FliS
Regulator biosynthesis protein FliO
Regulator biosynthesis protein FliN
Regulator biosynthesis protein FliH
Regulator biosynthesis protein FliF
Regulator biosynthesis and signal factor FlgR
Regulator basal-body rod protein FlgG
Regulator basal-body rod protein FlgG
Regulator basal-body rod protein FlgC
Regulator basal-body rod protein FlgB
Regulator basal-body rod protein FlgB
Regulator basal body P-ring protein FlgI
Regulator assembly protein FliH
Regulator assembly protein FliH
Regulator assembly peptidoglycan hydrolase FlgJ
Flagellar synthesis protein
Flagella protein
Flagella protein
Negative biosynthesis regulatory protein FlbD
Flagella basal body P-ring formation protein FlgA
Fimbrial usher protein
Fimbrial usher protein
Fimbrial domain-containing protein
Fibronectin/ peripheric chaperone
fatty acid metabolism regulator protein
fatty acid metabolism transcriptional regulator FadR
exopolysaccharide/ester
Exonuclease ABC subunit A
exonuclease ABC subunit A
Enhanced entry-like sensitivity protein SsuM
Endonuclease/exonuclease/phosphatase family protein
Endonuclease/exonuclease/phosphatase family protein
Elongation factor P hydroxylase EpmC (EF-P beta-lysylation pathway)
ElyPA/4 domain-containing protein
ElyPA/4 domain-containing protein
ElyPA/873 domain-containing protein
ElyPA/852 domain-containing protein
ElyPA/848 domain-containing protein
ElyPA/848 domain-containing protein
ElyPA/828 domain-containing protein
ElyPA/827 domain-containing protein
ElyPA/747 domain-containing protein
ElyPA/583 domain-containing protein
ElyPA/574 domain-containing protein
ElyPA/561 domain-containing protein
ElyPA/560 domain-containing protein
ElyPA/559 domain-containing protein
ElyPA/558 domain-containing protein
ElyPA/557 domain-containing protein
ElyPA/556 domain-containing protein
ElyPA/555 domain-containing protein
ElyPA/554 domain-containing protein
ElyPA/553 domain-containing protein
ElyPA/552 domain-containing protein
ElyPA/551 domain-containing protein
ElyPA/550 domain-containing protein
ElyPA/549 domain-containing protein
ElyPA/548 domain-containing protein
ElyPA/779 domain-containing protein
ElyPA/778 domain-containing protein
ElyPA/1035 domain-containing protein
ElyPA/1034 domain-containing protein
DNA-directed RNA polymerase specialized sigma subunit
DNA-binding transcriptional regulator LuxR-/FixK family
DNA-binding transcriptional regulator LuxR_Sun5 family
DNA-binding transcriptional regulator LuxR_Sun5 family
DNA-binding transcriptional regulator CsoS1
DNA-binding response regulator, SirepR family, contains REC and winged-helix (wHTH) domains
DNA-binding response regulator, RstA-Pho family, contains REC and winged-helix (wHTH) domains
DNA topoisomerase IA
DNA topoisomerase (ATP-hydrolyzing) subunit A
DMSO reductase anchor subunit DmsC
DMSO reductase anchor subunit DmsC
Divalent-cation tolerance protein Z1 type cytochrome biogenesis protein
disulfide bond formation protein SocB
dipeptide/tripeptide permease DppA
dihydrolipoyllysine-residue succinyltransferase
diaminopropionate isomerase subunit B
dihydroxyacetone
Dc- and tricarboxylate transporter
dipA family protein
D-serine/D-alanine/glycine transporter
D-lactate dehydrogenase
D-amino-acid transaminase
D-amino-acid dehydrogenase
cytochrome c-type protein
cysteine synthase CysM
cysteine synthase A
Cupin domain protein related to quercetin dioxygenase
CysK protein
cutinic-5 production protein
cutinic-E3 family cytochrome
Cofac-branching enzyme GlgB family cytochrome
CobL/cytochrome
Chaperone
class II fructose-bisphosphate aldolase
Class I SAM-dependent methyltransferase
class I ribonucleotide reductase maintenance protein YfaE
class I fructose-bisphosphate aldolase
chromosome segregation protein SmcB
chromosome segregation regulator ChrR
Chemotactic response regulator CheB, contains REC and protein-glutamate methylesterase domains
chemotactic protein CheA
chemotactic protein CheA
Cell division protein ZipA
cell division protein FtsK
Cell division inhibitor, MinD/ParA/Soj protein
CDP-diacylglycerol--glycerol-3-phosphate 3-phosphatidyltransferase
Cardiolipin synthase
cardiolipin/phosphatidylserine SynT
branched-chain amino acid transport system II carrier protein
branched-chain amino acid ABC transporter permease LivM
bifunctional tRNA (adenosine(37)-C(2)-)-methyltransferase TrmA/ribosomal RNA large subunit methyltransferase RlmN
bifunctional tRNA (5-methylaminomethyl-2-thiouridine)(34)-methyltransferase MnmC/FAD-dependent 5-carboxymethylaminomethyl-2-thiouridine(34) oxidoreductase MnmC
bifunctional 4-deoxy-4-formamido-L-arabinose-phosphoundecaprenol deformylase/4-amino-4-deoxy-L-arabinose-phosphoundecaprenol transferase ArnA
bifunctional acetaldehyde-CoA/alcohol dehydrogenase
bifunctional 2-polyprenyl-6-hydroxyphenol methylase/3-demethylubiquinol-10 methyltransferase UbiG
beta-ketoacyl-ACP synthase I
Beta-lactamase domain-containing protein
Bacteriocin-2 domain-containing protein
Basal-body rod modification protein FlgD
Autoinducer-2 production protein LsrG
ATP-binding protein
Aspartate/methionine/tyrosine aminotransferase
Aspartate aminotransferase
aspartate kinase
anaerobic reductase (glycine/sarcosine)
anaerobic reductase (glycine/sarcosine)
AraC-type DNA-binding domain and AraC-containing proteins
AraC family transcriptional regulator
AraC family transcriptional regulator
Anoxibac Sun1053 imelysous subunit
Anaerobic sclerocytokine-containing dehydrogenase
anaerobic sclerocytokine-containing dehydrogenase
aminoglycoside-N(3)-acetyltransferase
aminoglycoside N-acetyltransferase AAC(3)-IVa
aminodeoxychorismate synthase component I
aminophosphonobacterium/transferase
alpha-aminobutyrate/alanine/glycine transporter
Alpha-aminobutyrate family binary toxin subunit A
Alpha-galactosidase
Alpha-galactosidase
alpA family phage regulatory protein
alanine racemase
Acyl-CoA hydrolase
Acyl-CoA hydrolase
Acetylhydrolase
acetyl-CoA carboxylase, carboxyltransferase component
acetyl-CoA C-acetyltransferase FadA
acetate kinase
ABC type sulfide transport system, periplasmic component
ABC type dipeptide/oligopeptide/nickel transport system, permease component
ABC-type dipeptide/oligopeptide/nickel transport system, ATPase component
ABC transporter substrate-binding protein
ABC transporter substrate-binding protein
ABC transporter ATPase
ABC transporter ATP-binding protein
A/G-rhosomal protein L7/L12 group
A/G-rhosomal protein L7/L12 group
5'-deoxynucleotidase
4-phosphoserol-thronate dehydrogenase PdxB
4-hydroxy-tetrahydrodipicolinate synthase
3-oxoacyl-[4-acyl-carrier-protein] synthase III
2-succinyl-6-hydroxy-2,4-cyclohexadiene-1-carboxylate synthase
2-succinyl-5-enolpyruvyl-6-hydroxy-3-cyclohexene-1-carboxylic-acid synthase
1-acyl-sn-glycerol-3-phosphate acyltransferase
Yrdg domain-containing protein
WYL domain-containing protein
Tyr recombinase domain-containing protein
TypeA/BipA-subfamily bystrin GtpA family toxin
trimethoprim-resistant dihydrofolate reductase DfrA16
Thiol:disulfide interchange protein
Transposase
Transposase
Translational error-prone DNA polymerase V autoproteolytic subunit
Transcriptional regulator, MerR family
Transcriptional regulator, MerR family
Threonine dehydrogenase or related Zn-dependent dehydrogenase
TGc domain-containing protein
tetracycline resistance MFS efflux pump
tetracycline-inactivating monooxygenase TetX2
tetracycline efflux MFS transporter TetG5
tetracycline efflux MFS transporter TetG5
sulfonamide-resistant dihydropteroate synthase Sul3
sulfonamide-resistant dihydropteroate synthase Sul2
sulfonamide-resistant dihydropteroate synthase Sul1
Sbcde_like domain-containing protein
site-specific DNA-methyltransferase (adenine-specific)
Sfiba-specific DNA-recombinase SpoIVCA/SKI family
Mrcat1 chain dehydrogenase
Retron-type reverse transcriptase
Ribonuclease/invertase-type recombinase catalytic domain-containing protein
Resolvase
RspB family plasmid replication initiator protein
quaternary ammonium compound efflux SMR transporter QacE
proline-aldehyde oxymorphogenase-like domain-containing protein
PSE family antimicrobial-hydrolyzing toxin, A beta-lactamase CARB-2
Phosphotransferase
Phage integrase family
PadR family transcriptional
Orna protein
Orp
NAD-dependent aldehyde dehydrogenase
Mutator family transposase
mRNA interferase RelE
MDTA/NorM family multidrug efflux RND transporter
macrolide efflux MFS transporter MefE/Mel
LysR family transcriptional
IS21-like element ISAba3 family transposase
ISH family transposase
ISH family transposase
ISEfa family transposase
IS3 family IS1000 transposase
IS3 family transposase ORF A
IS3 family transposase ORF A
IS3 family ISMsm4 transposase ORF A
IS3 family ISMsm4 transposase ORF A
IS3 family ISAba14 transposase
IS256 family transposase
IS200/IS605 family element transposase ORF A
Integron integrase
Integron integrase
hypothetical protein
HTH-21-type domain-containing protein
HTH cro/C1-type domain-containing protein
HNH domain-containing protein
HNH domain-containing protein
Heavy-metal-associated domain-containing protein
GMP synthase (glutamine-hydrolyzing)
Error-prone DNA polymerase V autoproteolytic subunit
Efflux transporter periplasmic adaptor subunit
EF-hand domain-containing protein
EF-hand domain-containing protein
ElyPA/529 domain-containing protein
ElyPA/528 domain-containing protein
ElyPA/511 domain-containing protein
ElyPA/510 domain-containing protein
ElyPA/509 domain-containing protein
ElyPA/319 domain-containing protein
DNA methylase
Diaphragma-Cre4 domain/prolific folding accessory domain-containing protein
Cupin domain protein
CsoS1/CsoS1 family heavy metal efflux RND transporter
CopG family transcriptional regulator
Cobalt-zinc-cadmium resistance protein CzcA Cation diffusion facilitator family transporter
Cation/ taxate peptide
Cbs/adenylyltransferase
chloramphenicol efflux MFS transporter
Carbon transporter
BRO3-1 system phosphatase PapP type A
BRO3-1 system Lon protease-like protein Bro1
BRO3 system P-loop protein Ro3
BRO3 system Lon protease-like protein Bro1
ATP-binding protein
Antitoxin
AAA/ATP-lik family aminoglycoside nucleotidyltransferase ApdA3
aminoglycoside 3'-phosphotransferase APH(3')-IIIa
Ankyrin domain-containing protein
A-carboxyltransmembrane decarboxylase
Status
Multi-copy
Present
Absent
NT6-2_contig_13
3-1TC-A_contig_9
3-1TC-A_contig_2
NT3-1_contig_1
3-1TC-A (PP10+O2)_contig_9
Samples